

# WASP-11b

R-0712

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-10\_241119 · created 2026-07-27T22:56:59+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460633.7587 +/- 0.0015 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1415 +/- 0.0062               |
| Transit depth (Rp/Rs)^2                           | 2.0 +/- 0.18 [%]                |
| Orbital Inclination (inc)                         | 89.96 +/- 0.39                  |
| Airmass coefficient 1 (a1)                        | 1.1561 +/- 0.0077               |
| Airmass coefficient 2 (a2)                        | -0.134 +/- 0.022                |
| Scatter in the residuals of the lightcurve fit is | 0.67 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 3.57                            |
| Optimal Annulus                                   | 12.01                           |
| Transit Duration (day)                            | 0.10801 +/- 0.00072             |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.1415 ± 0.0062 vs 0.131 (deviation 1.69σ) |
| Duration fit vs published | 0.10801 d vs 0.1075 d (deviation 0.71σ)    |
| Mid-time O–C              | -1.2 min                                   |
| Depth SNR                 | 22.8                                       |
| Residual scatter          | 0.67 %                                     |

## Light curve

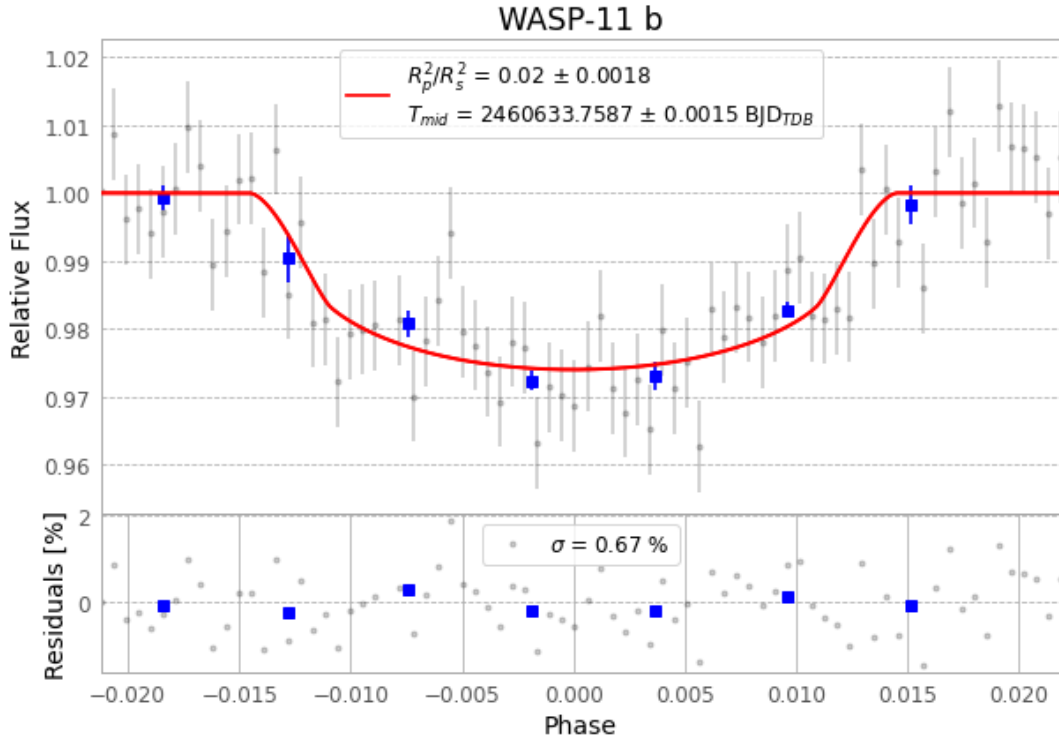


Figure 1 — FinalLightCurve\_WASP-11 b\_2024-11-18.png (SHA-256 d258dee9ab0ee788...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [414, 245], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 d6bb19315e65441f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 046045f

## Data provenance

78 FITS frames (51 MB), HATP-10241119040915.FITS → HATP-10241119080016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-241119.log (92b7a87a12a1...), FinalLightCurve\_WASP-11 b\_2024-11-18.png (d258dee9ab0e...), FinalLightCurve\_WASP-11 b\_2024-11-18.csv (7fabdf2d4a4b...)

## Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 22:57Z.